

Finding a Partnership Between Solar and Energy Efficiency

by Matt Golden

Coupling clean energy production technologies—such as solar electric and solar thermal—with efforts to reduce energy consumption in the home may seem like a natural pairing. But in reality, two individual industries have emerged. One is focused on conservation, the other on energy generation. Regulation, market forces, and good common sense demand that we reverse that trend and take a more holistic and systematic approach to improving home energy conservation (see “Why Are We Divided?” p. 26).

At Sustainable Spaces in San Francisco, California, we have been combining home performance with renewable energy systems to provide clients with comprehensive solutions to their energy needs since we opened our doors three years ago. Sustainable Spaces focuses on building performance and core system improvements, while partnering with solar installers in the San Francisco Bay Area to provide homeowners with a total solution that addresses energy, comfort, and health. Some solar installers call us to inspect every house they work on, while others call only when a client could clearly benefit from some energy efficiency advice, or when a client does not have a good solar location. One contractor calls only after a system is installed, so as not to complicate a sale that is in progress. We expect all of these partnerships to evolve and broaden, as new rebate regulations



ADAM WINTER

This Aprius evacuated-tube solar-thermal system (above) cuts natural gas consumption, while the 3.7 kW PV array offsets this efficient home's reduced electricity demand.

come into play with the passage of the California Solar Initiative (CSI).

Given the right circumstances, solar energy is beneficial both to homeowners and to society as a whole. However, California ratepayers, who subsidize these installations via a small tax on every utility bill, only receive the maximum benefit of their subsidies when a home's efficiency is maximized before a renewable energy system is sized and installed. New CSI regulations, which require that homeowners obtain energy efficiency audits in order to qualify for rebates, should ensure that ratepayers get the highest return on their investment (see “The California Solar Initiative,” p. 6).

While appropriately sized renewable energy solutions are part of many

of our projects, our experience at Sustainable Spaces has proved that the best way to create satisfied customers is first to evaluate each client's house as a system. This enables us to address our client's concerns regarding energy, comfort, and indoor air quality. Only after we have addressed such fundamentals as building envelope, insulation, and HVAC system do we generally recommend moving on to higher-tech solutions. Clients who choose a systems approach will see significant benefits in comfort, health, and safety—not just a reduction in their utility bills. While making a master bedroom comfortable or reducing condensation that is causing mold is a hard-to-quantify improvement in a return-on-investment calculation, changes like these provide

benefits that often outweigh simple energy savings.

TEAMWORK SUCCEEDS IN BERKELEY

We recently put our philosophy into action in a major remodeling project located in Berkeley, California. We started by modeling the building's performance, using our proprietary in-house software. This enabled us to assist the architect, contractor, and homeowner in making well-informed decisions on materials selection, system design, performance specifications, and renewable energy systems. The result was an opportunity to design the most comfortable and efficient whole-house system possible.

We first turned our attention to the fundamentals of insulation and envelope. Where more insulation was needed, we worked with the contractor to select green materials, such as batt insulation made from discarded blue jeans, and we set standards for installation. Thanks to our efforts, the calculated heating load was significantly less than it is for most buildings of the same size. We were also able to forego air conditioning, thanks to Berkeley's mild climate.

We designed the mechanical system based on two hydronic air handlers. A Lifebreath system with a built-in heat recovery ventilator (HRV) and a highly efficient electronically commutated motor (ECM) conditions the core of the house, and a First Co. (First Operations LP) air handler conditions the upper floor. Two systems were needed in this remodel because there was no easy way to run ducts from the ground floor to the top floor. Instead, we simply ran insulated cross-linked polyethylene tubing, or PEX, up the wall and installed a sec-

ond air handler. Hydronic forced air functions in essentially the same way as a traditional forced-air furnace, except that it uses a high-efficiency water coil instead of a gas-fired heat exchanger. The Lifebreath system provides filtered ventilation air to the house, either on a programmed schedule or on an as-needed basis. The built-in HRV allows this system to recover as much as 80% of the energy from the exhausted air flow.

A hydronic forced-air system has a number of advantages over

combustion process can take place outside the living space, reducing the risk of CO poisoning. The air handler is a separate unit, which can be located wherever it is most convenient.

Because both of these units are designed to use potable water, we were able to power both domestic hot water and space-heating needs with a single combustion source. In this case we used an existing water heater that had an 80% recovery efficiency, which had been installed

BY TAKING A SYSTEMS APPROACH TO A HOUSE FROM THE OUTSET, WE WERE ABLE TO ENGINEER A SOLUTION THAT COUPLED ENERGY EFFICIENCY AND LOAD REDUCTION WITH APPROPRIATELY SIZED RENEWABLE ENERGY SYSTEMS.

traditional gas-fired forced air. One advantage is that the Btu output can easily be adjusted by reducing pump speed or water temperature. This is particularly important in the Bay Area, where many homes have loads that are significantly lower than the output of the smallest gas furnace on the market. In this case, because of the need to install two units, we were able to adjust the flow of each unit to match the actual load of the rooms it served, calculated using *Manual J*. These systems run at heat exchange temperatures that are 80% lower than those of the typical gas furnace. Water enters the air handler at about 135°F and leaves at about 95°F, eliminating the scorched-air smell associated with traditional forced air. Finally, because the water heater can be located in the garage or on the side of the house, the

just before we were hired on to the project. However, this system can also be designed with a high-efficiency tanked system, a boiler, or even a tankless unit.

To reduce the amount of natural gas consumed, we installed a 90-tube Apricus evacuated-tube solar-thermal water-heating system. This system captures solar radiation at an efficiency rate of 66% and converts it into hot water. The solar-heated water is stored in a superinsulated 200-gallon Trendsetter Industries water tank and is used to preheat the incoming water. City water, as well as the return water coming from the air handlers, passes through a heat exchange loop in this tank before entering the water heater, dramatically reducing gas consumption.

Evacuated tubes operate particularly well for space conditioning,

WHY ARE WE DIVIDED?

While solar power has many benefits, most solar-rebate structures create a conflict among the interests of utilities, solar contractors, the home performance industry, and homeowners. On the macro level, the subsidies are intended to reduce peak loads and overall demand on the energy grid. However, the rebates that were put in place to stimulate the creation of the now-booming California solar industry are clashing with the other objective of the subsidies—reducing overall demand and shaving peak loads.

Both utility companies and the California Public Utilities Commission (CPUC) explicitly state that energy efficiency is the cheapest and cleanest way to gain additional capacity. However, traditional rebate structures entice effective PV technologies to offset residential energy consumption, with no incentive for energy efficiency. Installing a large solar system on a home that is otherwise energy inefficient does not serve the broader purpose of the incentive. If, by implementing efficiency measures, the necessary size of a solar system can be reduced from 5 kW to 3 kW, the state will retain as much as \$5,000 in rebate funds that can be leveraged for additional projects, while achieving the same positive outcome in terms of energy footprint. At \$2.50 per watt, issuing rebates is a costly measure that targets only part of the equation.

Rebates also tend to skew head-to-head return-on-investment (ROI) comparisons. Energy efficiency measures, for which there are no significant public rebates, often can't compete in terms of straight ROI with heavily subsidized technologies such as PV, even though they may do more to reduce energy use. In fact, in California, where typical electric rate schedules increase kWh costs based on consumption, whichever system is applied first generates the best return on investment. Reduce a bill by 300 kWh a month and a large consumer in a top tier paying \$0.35 per incremental kWh will save \$100 per month. But reduce that same consumer's bill by another 300 kWh a month, and that consumer will save only \$45 per month, having been kicked



This LifeBreath furnace includes an HRV and an electronically commutated motor (ECM).

down into a lower tier with a rate of \$0.15/kWh (see Table A). Fortunately, with the passage of the California Solar Initiative, the way that rebates are structured is changing (see “The California Solar Initiative,” p. 6).

At present, there is a built-in tension between solar companies and the home performance industry. This tension has impeded our ability to work together. First, there is the perception—and occasionally even the reality—that home performance and solar contractors are competing for the same home improvement dollars. “Do I fix my furnace and insulation or do I install a 4 kW solar system?” a homeowner might wonder. In our experience, most homeowners, with some education, can be brought to understand that approaching this question from an either/or perspective is not useful. A holistic approach to improving overall home performance will lead the homeowner to improve energy efficiency and at the same time to install a renewable energy system—albeit perhaps a smaller one.

This approach, however, leads directly to the second impediment to greater cooperation. Businesses that sell solar systems profit from the installation of systems—and frequently, the bigger the system, the

greater the profit. A seller of solar who is compensated based on system size has no incentive to suggest turning off that old refrigerator in the garage—full of “emergency” beer—since to do so might reduce the amount of solar required by 1kW, saving the client and the state ratepayers thousands of dollars. When a home performance contractor comes in and reduces a customer's energy demand by 35%, that customer will purchase a smaller system. Still, for a solar installer, it can be more profitable to sell an appropriately sized system coupled with efficiency than to try to sell a larger system and have to compete on cost alone with other solar installers. If everyone is offering to sell a 6 kW system, offering to install a 4 kW

Table A. Rate Tiers for PGE e-1

Baseline	101-130%	131-200%	201-300%	300%+
\$0.11	\$0.13	\$0.2198	\$0.3029	\$0.3465

system plus insulation and duct sealing can give a company a competitive edge, while avoiding a head-to-head dogfight over cost per kilowatt.

With competition increasing in the solar industry, changes in regulation, and a public that is becoming increasingly educated about energy use, it is time to overcome the barriers that keep our industries apart and find new, smarter ways to work for our clients and toward our shared goals. As evidenced through successful partnerships that we have forged with solar installers in the San Francisco Bay Area, there is desire on the front lines to work through these issues.

Fundamental changes in the California solar rebate program will soon require all participants to take a more holistic view of home energy. With the state establishing a rebate system that more accurately targets the goal of creating a low-energy footprint, rather than simply subsidizing products, we may start to see more of an integrated approach to energy production and efficiency—an approach that offers the greatest rewards, both for the homeowner and for the public good.



Matt Golden installs an HRV heat exchanger core.



Evacuated tubes work efficiently at deflected angles and under low light.

compared to traditional flat-plate collectors. They are effective even on very cold days and in low light, heating water to high temperatures ranging from 125°F on cold winter days to 165°F in the summer (based on Bay Area insolation levels). Excess heat generated in summer months will be used to heat a lap pool that is soon to be installed.

In addition to the solar-thermal system, a grid-tied, 3.7 kW PV array was installed to offset electrical load. This array, consisting of Sun Power modules that were installed by Marin Solar, will offset an estimated 60% of electrical consumption for this home.

By taking a systems approach to a house from the outset, we were able to engineer a solution that coupled

energy efficiency and load reduction with appropriately sized and properly designed renewable energy systems. By approaching the problem holistically rather than with a piecemeal combination of various trades, we achieved much better results for our client.

BRIDGING THE GAP

With advanced testing tools and the body of building science that is readily available, the home improvement industry is poised to move beyond the single-trade and -product paradigm. It has become irrefutable that a well-engineered whole-house system, rather than any particular product, is the key to high performance in a home.

Sustainable Spaces works with each client to quantitatively measure a home's performance,

creating a detailed diagnosis of existing systems and a prioritized step-by-step prescription. Once a client decides to move forward, we bring in specially trained contractors to conduct the necessary remediation steps, based on specific performance specifications. Sustainable Spaces coordinates the project, and provides quality assurance by testing out and inspecting the work.

For many people, a building science approach to home improvement can be easier to understand if you compare it to an auto mechanic's approach to fixing a car. Whom would you prefer doing business with—the mechanic who takes an hour to diagnose your car's problem and then fixes it once, or the mechanic who

keeps replacing parts until eventually the engine turns over? Which mechanic would you recommend to your friends?

Although it is fun to design and install advanced technologies, most clients are better served by a focus on their home's fundamentals—building envelope, HVAC system, duct leakage, moisture management, water consumption, lighting, and appliances. Most of our jobs consist of simple technologies, designed and installed properly, in the context of the entire system. Once homeowners understand their house as a system, they will almost certainly take steps to improve their home's performance.

Aside from serving the best interests of the client, this approach offers other benefits. It fosters broad cooperation and partnership between energy efficiency and renewable energy providers. Progress in this direction is inevitable; market forces, changes in regulations, and client demands are rapidly moving us toward this new reality. From lightbulbs and appliances to HVAC and solar power, it is all one home energy system. Working together, we can create solutions that are better for our clients and contribute to a greater good on a global scale.



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